IOT PROJECT

Initial Project Proposal

1. Group Information

Provide the following details for every member of your group:

#	Full Name	Registration Number	Email / Contact
1	Adam Wahid	231027074	adamw5977@gmail.com
2	Ahmed Mohamed	231027052	amkzaher@gmail.com
3	Ahmed Nabil	231017825	ahmadnabil20001@gmail.com

2. Communication Protocol(s)

Specify the messaging or communication protocol(s) your system will use for IoT data transmission. Justify your choice briefly if possible.

– Selected protocol(s) *

e.g., MQTT, AMQP, CoAP, HTTP/REST, WebSocket, etc.

MQTT

We chose MQTT because it is lightweight and supports real-time communication using publish/subscribe, making it ideal for IoT systems.

3. User Interface (GUI) Platform

State whether your graphical user interface will be a Mobile Application or a Web Application, then provide the relevant technology stack.

3a. Platform Choice

– Mobile Application –OR– Web Application *

Web application

3b. If Mobile Application

– Programming language(s) and framework(s) *

e.g., Flutter (Dart), React Native (JavaScript), Kotlin (Android), Swift (iOS), Ionic, etc.

3c. If Web Application

– Programming language(s) and framework(s) *
  e.g., JavaScript / React, JavaScript / Vue, Python / Django, Python / Flask, Node.js / Express, etc.

JS,HTML,CSS,Python/Flask

4. Physical Maquette / Prototype

Describe the physical model or prototype your project will simulate or control.

– Type of maquette *
  e.g., Water tank, toy car, custom-made wooden house, greenhouse model, conveyor belt, etc.

3d printed cradle

5. Electronic Components

List all hardware components you plan to use, including microcontrollers, sensors, actuators, and connectivity modules.

– Microcontroller / development board(s) *
  e.g., Arduino Uno, Arduino Nano, ATmega328, ESP8266, ESP32, Raspberry Pi, etc.

Arduino uno

– Sensor(s) *
  e.g., Water level sensor, temperature/humidity sensor (DHT11/DHT22), ultrasonic sensor, weight sensor (HX711 + load cell), PIR motion sensor, etc.

sound sensor

– Actuator(s) and output components *
  e.g., Relay module, servo/DC motor, LED strip, buzzer, LCD/OLED display, etc.

servo motor

– Connectivity / communication modules *
  e.g., ESP Wi-Fi module, NRF24L01 (RF), SIM800L (GSM), Bluetooth HC-05, etc.

ESP Wi-Fi module

6. Additional Information

Include anything else that is relevant to your project concept, such as:

– Cloud or back-end platform
  e.g., Firebase, AWS IoT, Azure IoT Hub, self-hosted MQTT broker, etc.

Mosquitto (manual MQTT broker)

– Database solution
  e.g., InfluxDB, Firebase Realtime DB, PostgreSQL, MongoDB, etc.

SQL lite

– Security or authentication considerations

Log in system

– Any notable technical challenges or constraints you anticipate

7. Task Assignment (Optional but Recommended)

Optional – Assign each member a primary area of responsibility. Every member is still expected to understand the fundamentals of their teammates' work and be able to discuss them during the in-class review.

Group Member	Primary Task / Area	Secondary Familiarity

> ⚠ **Reminder**
> At least one member from each group will be asked to present and discuss these points directly with the instructor. All members should be familiar with every section of this submission.

* Required field